

Jahangirnagar University
Department of Computer Science and Engineering
3rd Year 2nd Semester
CSE 357 (Microprocessors)

Time : 40 minutes

Total Marks:15

Answer any three of the following questions

1. Discuss the evolution of microprocessors. 5
2. What are the basic differences between an accumulator based microprocessor and a general purpose microprocessor? Define RISC microprocessor. (2.5+2.5)
3. Differentiate between RISC and CISC type microprocessor. Discuss PUSH and POP operation of microprocessor. (2.5+2.5)
4. Discuss the FLAG register of 16 bit microprocessor. 5
5. Draw the pin diagram of 8086 microprocessor when it works in maximum mode and give a truth table of pin no. 26,27 and 28. 5

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Tutorial no.2

Time : 40 minutes

Total Marks:10

Answer any two of the following questions

1. Discuss the READ and WRITE cycle timing diagram of a typical semiconductor memory with different parameters. 5
2. Define and state the purpose of interrupts. Draw an interrupt vector table. 5
3. What do you mean by addressing modes? Explain at least six addressing modes of 8086 microprocessor with examples? 5

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Answer any three of the following questions

1. Describe briefly synchronous and asynchronous communication.
2. Illustrate how a 64 key keyboard and an 8 digit seven segment display can be connected to an 8279.
3. Define 2- key lockout and N-key rollover. Draw the general structure of the Intel 8279 Keyboard/display controller.
4. Draw and describe internal architecture of 80286 microprocessor. What are program invisible registers? 5
5. Write a short note on memory addressing modes of 80386. 5

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4. What is HyperThreading Technology. Draw and describe internal architecture of Pentium processor. 5
5. Write down the differences between ~~i~~ Dual Core and Quad Core Processor.
ii) Intel Core i3, i5 and i7 Processor
6. i) Write down the features of Pentium processor.
ii) compare between Pentium and 80386 processor.

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Answer any three of the following questions

1. Differentiate between RISC and CISC type microprocessor. Discuss PUSH and POP operation of microprocessor. (2.5+2.5) 5
2. Discuss the FLAG register of 16 bit microprocessor. 5
3. Draw the internal architecture of 8086 microprocessor. Explain how physical address is calculated from logical address in a 16 bit microprocessor. (3+2) 5
4. Describe main memory array design using full decoding technique. 5
5. Draw the block diagram of 8237 DMA controller along with associated logic and explain its role as a slave and master. 5
6. Discuss the READ and WRITE cycle timing diagram of a typical semiconductor memory with different parameters. 5
7. What is the difference between the 80286 real address mode and protected virtual address mode? Explain how protected virtual address mode is used to calculate the physical address. 5
8. Draw and describe internal architecture of 80286 microprocessor. What are program invisible registers? 5
9. Write a short note on memory addressing modes of 80386. 5